Retail Purchase Behavior Study

1. Overview of the Project

This study examines how customers shop by looking at 3,900 individual transactions spanning a range of product categories. The purpose is to surface patterns in spending, identify distinct customer groups, understand which products perform best, and evaluate subscription habits so the business can make informed strategic choices.

2. Overview of the Dataset

The dataset consists of 3,900 records spread across 18 fields. It brings together three broad categories of information:

- Customer profile details, such as age, gender, location, and whether they hold a subscription
- Order specifics, including the item bought, its category, the amount spent, the season, size, and color
- Behavioral signals, such as discount and promo code usage, prior order count, how often the customer buys, review scores, and delivery method

Note: the Review Rating field is missing 37 entries.

- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

3. Preparing and Exploring the Data with Python

The initial phase focused on cleaning and getting familiar with the data:

- **Loading the data:** the dataset was brought into a pandas DataFrame.
- **First look:** df.info() was used to inspect the structure, and .describe() produced summary statistics.
- **Handling gaps:** null values were flagged, and the missing Review Rating entries were filled in using each product category's median score.
- **Naming conventions:** column headers were converted to snake_case to make the dataset easier to read and document.
- **New fields:** an age_group column was built by bucketing ages into ranges, and a purchase_frequency_days column was derived from order timing.
- **Redundancy check:** discount_applied and promo_code_used were compared for overlap; promo_code_used was removed as duplicative.
- **Moving to a database:** the cleaned data was pushed from the Python script into a PostgreSQL database for further querying.

4. Querying the Data in SQL

A series of PostgreSQL queries were run to answer specific business questions.

4.1 Revenue split by gender — total sales generated by each gender group.

NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

4.2 Big spenders who still used discounts — customers who applied a discount code yet spent more than the overall average order value (839 matching rows in total; sample below).

Missing Data Handling: Checked for null values ar
`Review Rating` column using the median rating o

Column Standardization: Renamed columns to **sn**
documentation.

Feature Engineering:

- Created **age_group** column by binning custo
- Created **purchase_frequency_days** columr

Data Consistency Check: Verified if `discount_ap`
were redundant; dropped `promo_code_used`.

Database Integration: Connected Python script to I
DataFrame into the database for SQL analysis.

ata Analysis using SQL (Business Tran

erformed structured analysis in PostgreSQL to answer

. **Revenue by Gender** – Compared total revenue ger
customers.

4.3 Best-reviewed products — the five items with the strongest average customer rating.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

4.4 Standard vs. express delivery — comparing average order value by shipping choice.

spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64

4.5 Subscribers vs. non-subscribers — customer counts, average spend, and total revenue by subscription status.

3	4	90
4	7	85
5	9	97
6	12	68

4.6 Products most reliant on discounts — the five items with the highest share of discounted orders.

	item_purchased text	Average Product Ratir numeric
1	Gloves	
2	Sandals	

4.7 Customer tiers — customers grouped into New, Returning, and Loyal segments based on their purchase history.

4	Hat	3.80
5	Skirt	3.78

- **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

4.8 Leading items within each category — the three top sellers in every product category.

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

4.9 Frequent buyers and subscription uptake — checking whether customers with more than five prior purchases subscribe at a higher rate.

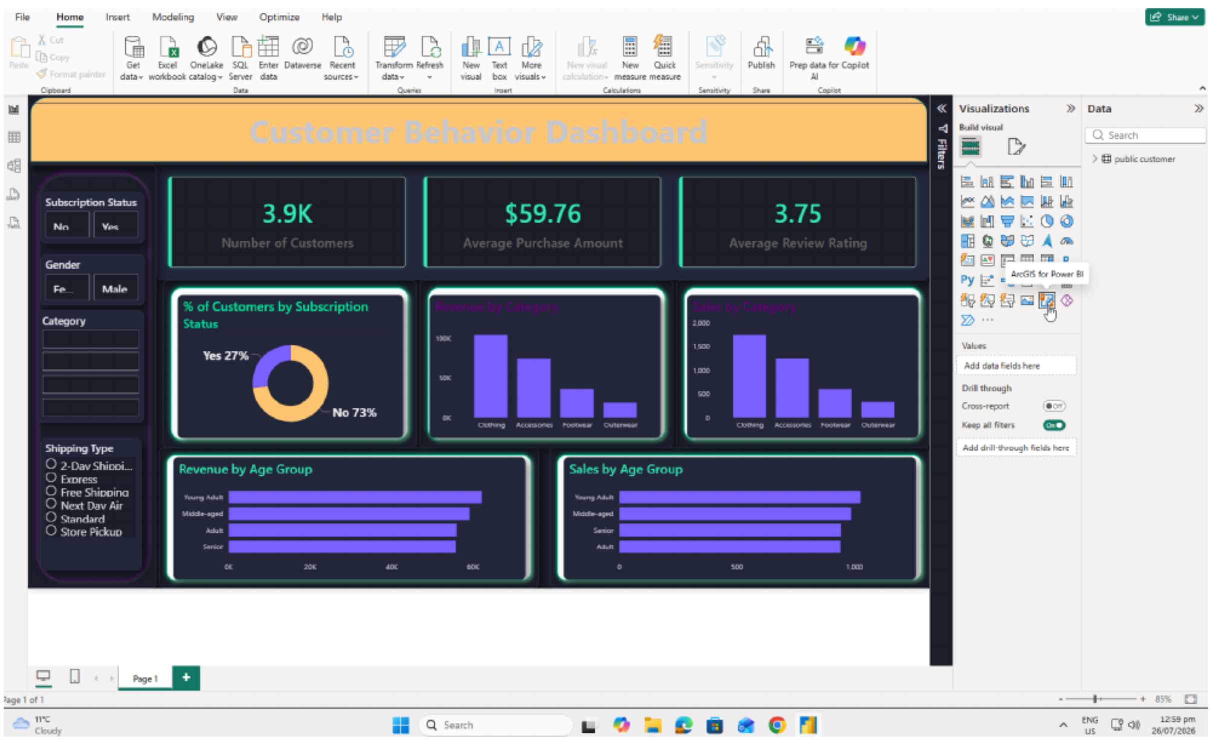
7. **Customer Segmentation** – Classified customers into New, Return segments based on purchase history.

	customer_segment text	Number of Custom bigint
1	Loyal	

4.10 Revenue contribution by age bracket — total sales attributable to each age group.

5. Visual Dashboard Built in Power BI

A Power BI dashboard was built to give stakeholders an at-a-glance visual summary of shopping trends. It brings together key metrics, a breakdown of subscribers, revenue and sales figures by category, insights split by age bracket, and filters that let users drill into the data interactively.



6. Recommendations for the Business

- **Grow the subscriber base:** offer members-only perks to encourage more sign-ups.
- **Reward loyalty:** give repeat shoppers incentives that help move them into the Loyal tier.
- **Reassess discounting:** weigh the sales lift from promotions against the impact on margins.
- **Feature strong performers:** spotlight highly rated, best-selling items in marketing campaigns.

- **Sharpen targeting:** concentrate marketing spend on the age groups that drive the most revenue and on customers who choose express shipping.